Moment of Inertia: Lab Analysis

SYNOPSIS

Note 3 asserted nothing and derived everything, and it ended at $\tau = I\alpha$. **The goal of this lab is to verify $\tau = I\alpha$** — not to use it.

The apparatus is a metal disc free to turn on an axle, with a small pulley of radius r_P on the same axle, a string wound round the pulley, and a hanging mass M_H on the other end of the string. Let go, and the falling mass winds the string off and spins the disc.

Two free-body diagrams run the whole analysis: one for the hanging mass, which gives you the tension, and one for the disc, which turns that tension into a torque. The string ties the two together — it cannot stretch, so $a = r_P\alpha$. Vary M_H, measure α each time, plot α against τ, and if $\tau = I\alpha$ is right the points lie on a straight line through the origin whose slope is $1/I$.

Then the check that makes it an experiment rather than a demonstration: compare the I you read off that slope with the $I = \frac{1}{2}M_D r_D^2$ you get from weighing the disc and measuring it with a ruler. Two independent routes to the same number.

BEFORE YOU READ ON — PREDICT

The hanging mass is falling. Is the tension in the string equal to its weight? Write one sentence saying yes or no, and why.

PERFORMANCE EXPECTATIONS

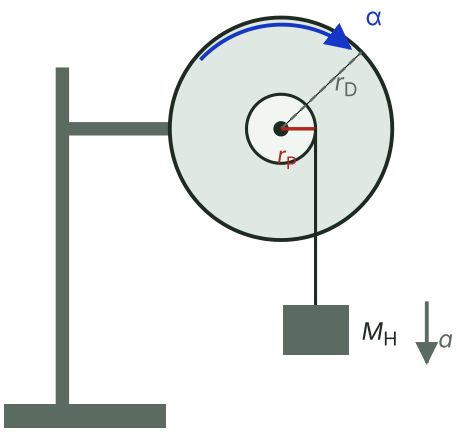
Draw the free-body diagram for each body in a connected rotational system and write Newton's second law for each. Use the inextensible-string condition $a = r\alpha$ to couple them. Predict the angular acceleration from the hanging mass. Linearise $\tau = I\alpha$ and get the moment of inertia from the slope of a graph. Compare an experimental value with an independently calculated one and account for the difference.

For PHY 151 — with calculus: Get $a = r_P\alpha$ by differentiating the string constraint rather than arguing it, and do the whole problem in one line from $\tau_{net} = d\boldsymbol{L}/dt$ for the system.

The apparatus, and the two free-body diagrams

Two bodies, one string. Treat them separately first and connect them afterwards — that is the whole method, and it is the same one you used for two blocks and a pulley back in Module 3.

pulley of radius r_P on the same axle



disc and pulley on one axle

the hanging mass **the disc**



Draw both free-body diagrams. On the hanging mass, every force on it. On the disc, only what can exert a torque about the axle — and on the lines underneath, say why its weight and the axle's reaction are not on your diagram.

1. Write Newton's second law for the hanging mass, taking down as positive, and rearrange it for the tension T.
2. Is T bigger or smaller than $M_H g$, and by how much? What would T be if the disc had no moment of inertia at all?
3. Write the torque the string exerts on the disc about the axle, and then Newton's second law for rotation. Why do the disc's weight and the axle's reaction force not appear?

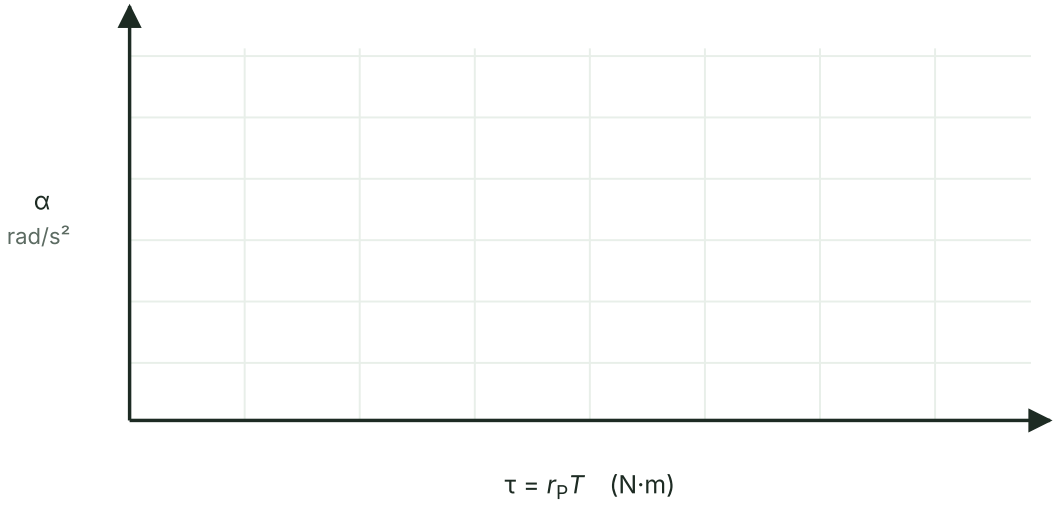
From the hanging mass to the torque

Two equations, three unknowns — a, T and α. The third equation is the string. It does not stretch, so the length that unwinds off the pulley is the distance the mass falls, and differentiating that statement twice ties the two accelerations together.

4. Write the condition that ties a to α, and combine it with your two equations to eliminate T and a. You should get a single equation in α and M_{H}.
5. Solve that for α. Check the two limits: what does your expression give as $M_{\mathrm{H}} \to 0$, and what does it give if $I \to 0$?
6. Take $r_{\mathrm{P}} = 0.025$ m, $I = 0.0030$ kg·m² and a hanging mass of 0.200 kg. Find (A) α, in rad/s²; (B) a, in m/s²; (C) T, in N; and (D) τ, in N·m. Then check (D) against $I\alpha$.

The graph, and what its slope is

Everything above *assumed* $\tau = I\alpha$. The experiment tests it, and the test is a straight line: vary the hanging mass, measure the angular acceleration each time, compute the torque from the tension, and see whether the points fall on a line through the origin.



Plot your five points and draw the best straight line through them. Mark the slope, write what it equals in terms of *I*, and then sketch in dashed what the line would look like if the axle had a constant friction torque.

7. You plot α on the vertical axis against τ on the horizontal. (A) What is the slope, in terms of I? (B) What should the intercept be? (C) If instead you plotted τ against α, what would the slope be?

8. Here are five runs with $r_P = 0.025$ m. Fill the third column from $\tau = r_P M_H (g - r_P \alpha)$, then get I from the data.

M_H (kg)	α measured (rad/s²)	τ (N·m)
0.050	4.04	
0.100	8.00	
0.150	11.88	
0.200	15.68	
0.250	19.41	

9. The disc itself has mass 1.50 kg and radius 0.064 m. (A) What is $I_D = \tfrac{1}{2} M_D r_D^2$? (B) By what percentage does it differ from the value you got from the slope? (C) Which of the two would you quote, and why?

What could make the two values disagree

They did disagree, by 2.3%. Every term you left out of the model is a candidate, and each one leaves a different fingerprint on the graph.

10. A constant friction torque τ_f at the axle opposes the motion. Redo the rotational equation with it included, solve for α, and say what it does to the slope and to the intercept of the α-against-τ plot.
11. Name two other things the model leaves out, and for each say whether it would make the I from the slope come out too big or too small.

FOR PHY 151 — WITH CALCULUS

Two steps above were argued rather than derived: the string condition, and the elimination. Both are one line with calculus, and the second line is worth knowing because it generalises to every connected rotational system you will meet.

12. The string does not stretch, so the length unwound is the distance fallen: $s = r_P\theta$. Differentiate it twice. Why is this, and not a force argument, what licenses $a = r_P\alpha$?
13. Write the total angular momentum of the system about the axle — the disc's $I\omega$ plus the hanging mass's $M_H v r_P$ — and apply $\tau_{net} = d\boldsymbol{L}/dt$ with $\tau_{net} = r_P M_H g$. How many lines does it take?

Looking ahead

This is the last of Module 5's teaching notes, and it is the one that closes the loop: Note 3 derived $\tau = I\alpha$ from $F = ma$ and the bridge relations, and this Note measures it. The sheets that remain in the module — the centripetal-force examples and the gravitation example — are practice on Notes 1 and 2 rather than new material.

Where this sits. The method generalises well past this lab. Any two bodies joined by a string over a pulley, any rolling body on an incline, any yo-yo: one free-body diagram per body, a constraint linking the accelerations, and eliminate the tension. The only thing that changes is which moment of inertia goes in.

BACK TO YOUR PREDICTION

No — the tension is *less* than the hanging mass's weight, by exactly $M_H a$. If the two were equal the net force on the mass would be zero and it would not accelerate at all, so the fact that it is falling faster and faster is itself the proof. In the worked numbers the weight is 1.96 N and the tension is 1.88 N: only 4% apart, which is exactly why the mistake is so easy to make and so hard to notice.

Check yourself

1. A uniform disc of mass 2.0 kg and radius 0.10 m turns about its centre. What is its moment of inertia, in kg·m²?

_____ kg·m²

2. While the mass is falling, the tension in the string is

- ○ equal to $M_H g$, because the string holds it
- ○ less than $M_H g$, because the mass is accelerating downward

- ○ greater than $M_H g$, because the disc resists turning
- ○ zero, because the mass is in free fall

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3. Plotting α against τ for this apparatus, the slope is

- ○ I
- ○ $1/I$
- ○ $r_P I$
- ○ r_P / I

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4. The string pulls with 1.88 N on a pulley of radius 0.025 m. What torque does it exert, in N·m?

_____ N·m

Your notes

BEFORE THE VIDEO

Guided notes — blank (PDF)

print or open on a tablet; record your prediction, then work the items as the video goes through them.

AFTER THE MODULE

Completed notes (PDF)

released when Module 5 closes.

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